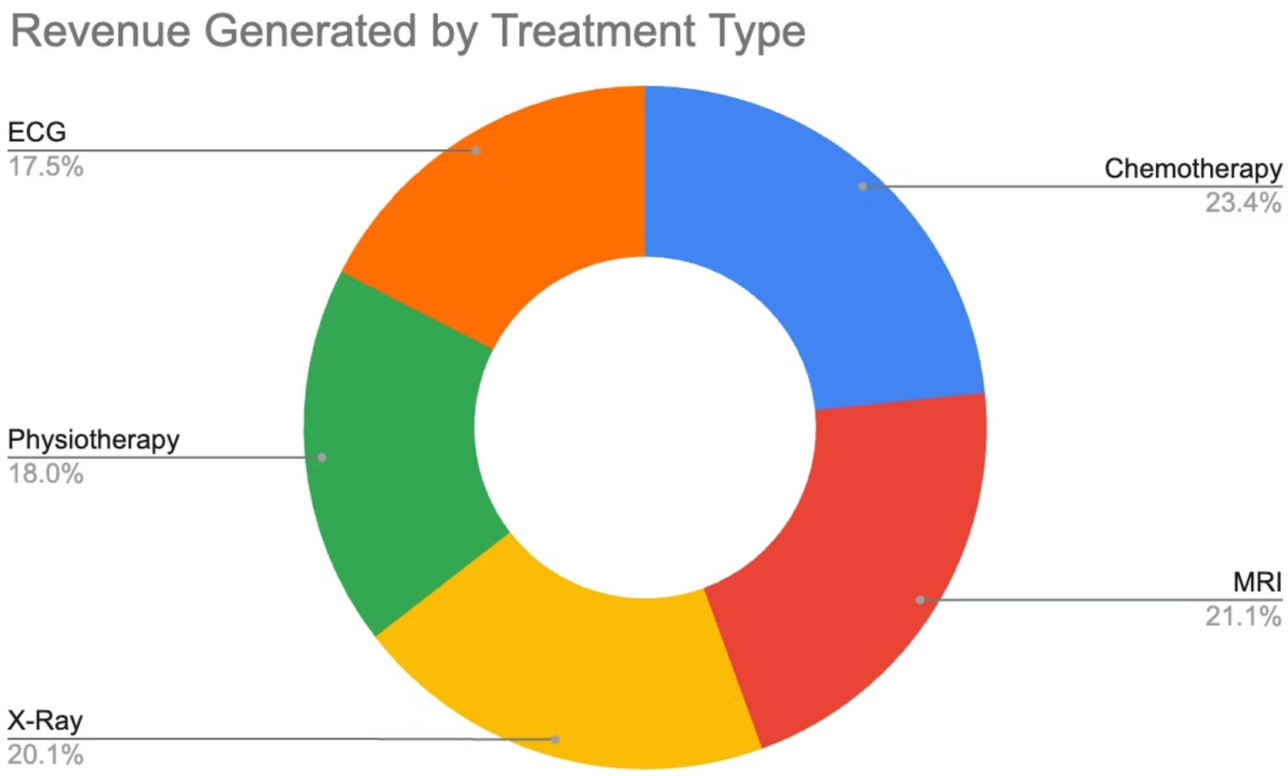
Claims Integrity and Risk Analysis

Prepared for: Health Insurance Claims Department and Analytics Teams

INSIGHT 1

Identifying High-Cost Treatment Areas



Understanding financial risks tied to various treatment types is crucial. Our analysis revealed that **Chemotherapy** incurred the highest overall cost across 49 appointments. Meanwhile, **MRI** had the highest average cost per appointment, from a total of 36.

This breakdown of treatment costs and their contribution to total revenue is vital. It allows us to scrutinise each treatment type to ensure the legitimacy of generated revenues and pinpoint potential areas of concern.

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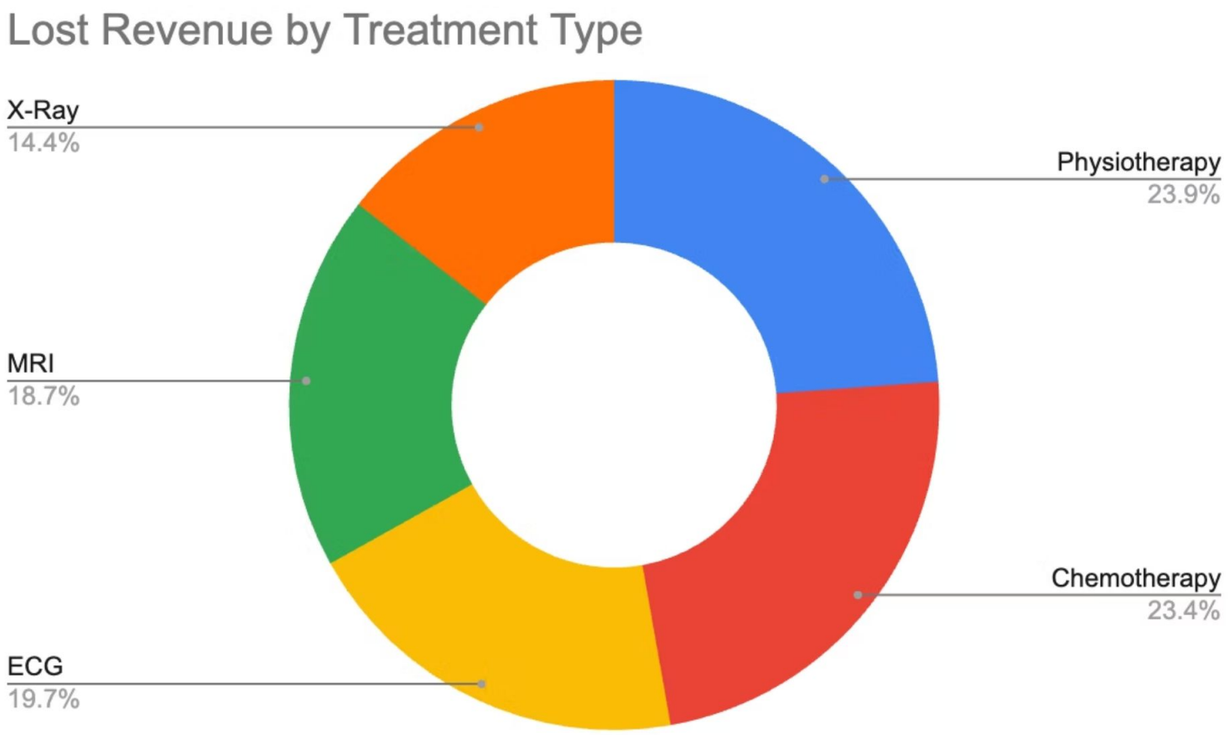
Revenue Loss from Failed Payments



We investigated the revenue lost due to claims with "Failed" payment statuses. The findings highlight **Physiotherapy** and **Chemotherapy** as the two treatment types with the most significant revenue loss.

This indicates a critical area for immediate attention, as these losses directly impact financial stability and operational efficiency. Further investigation into the root causes of these failed payments is essential.

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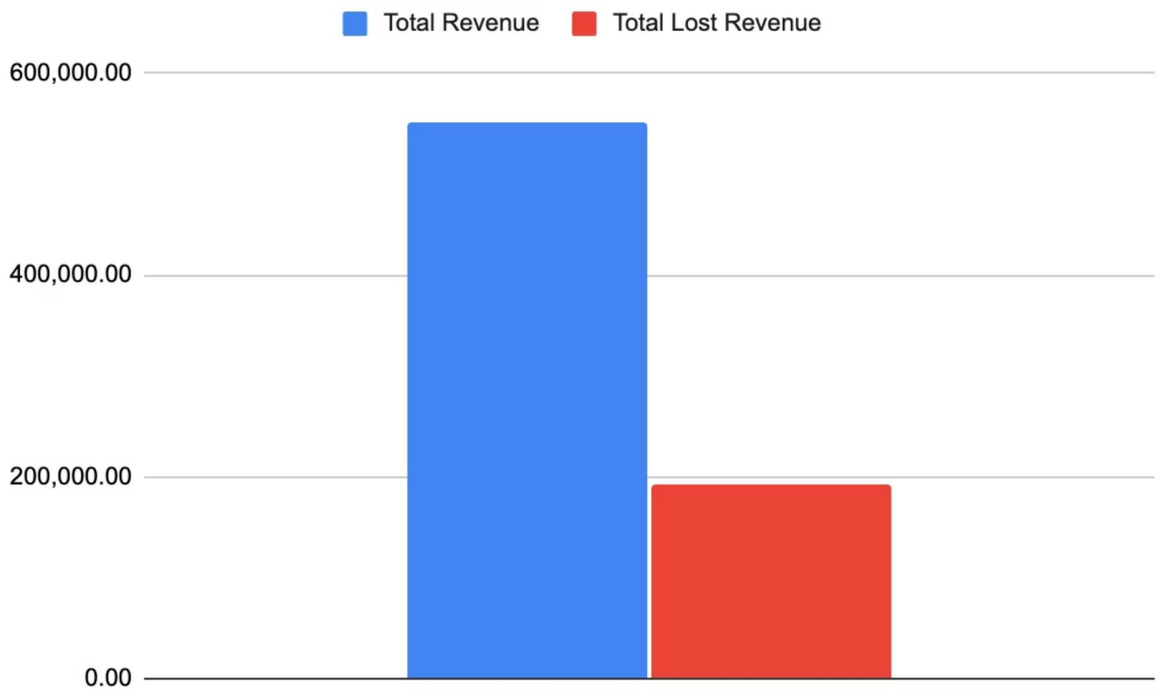
Significant Revenue Impact



Overall, our analysis shows that a substantial **35.05%** of total generated revenue was lost due to failed payments. This represents a critical financial challenge that demands immediate action.

The claims department must investigate the underlying reasons for these failed payments and implement effective strategies for recovery. This includes reviewing current payment processes and exploring alternative collection methods to mitigate future losses.

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INSIGHT 2

Analysing Claims Patterns and Accuracy

Verifying claims accuracy is paramount for several reasons:

- Enhances trust between patients and healthcare providers.
- Reduces risks of audits and penalties from regulatory bodies.
- Ensures compliance with legal medical standards.

Our review of appointment records revealed a concerning trend: out of 200 appointments, 103 (over 50%) were either cancelled or marked as no-shows. This indicates a significant inefficiency in scheduling and patient adherence.

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Discrepancies in Payment Status for Unfulfilled Appointments

Further investigation into unfulfilled appointments uncovered a critical billing error: 81 out of the 103 canceled or no-show appointments had their payments fulfilled.

Row	appointment_id	status	payment_status
1	A002	No-show	Paid
2	A003	Cancelled	Paid
3	A008	Cancelled	Paid
4	A014	Cancelled	Paid
5	A015	No-show	Paid
6	A019	Cancelled	Paid
7	A021	No-show	Paid
8	A022	No-show	Paid

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This indicates that the billing department approved payments for services that were never rendered, leading to significant financial waste and potential fraud. This highlights an urgent need to reconcile payment statuses with actual appointment completion to prevent further losses.

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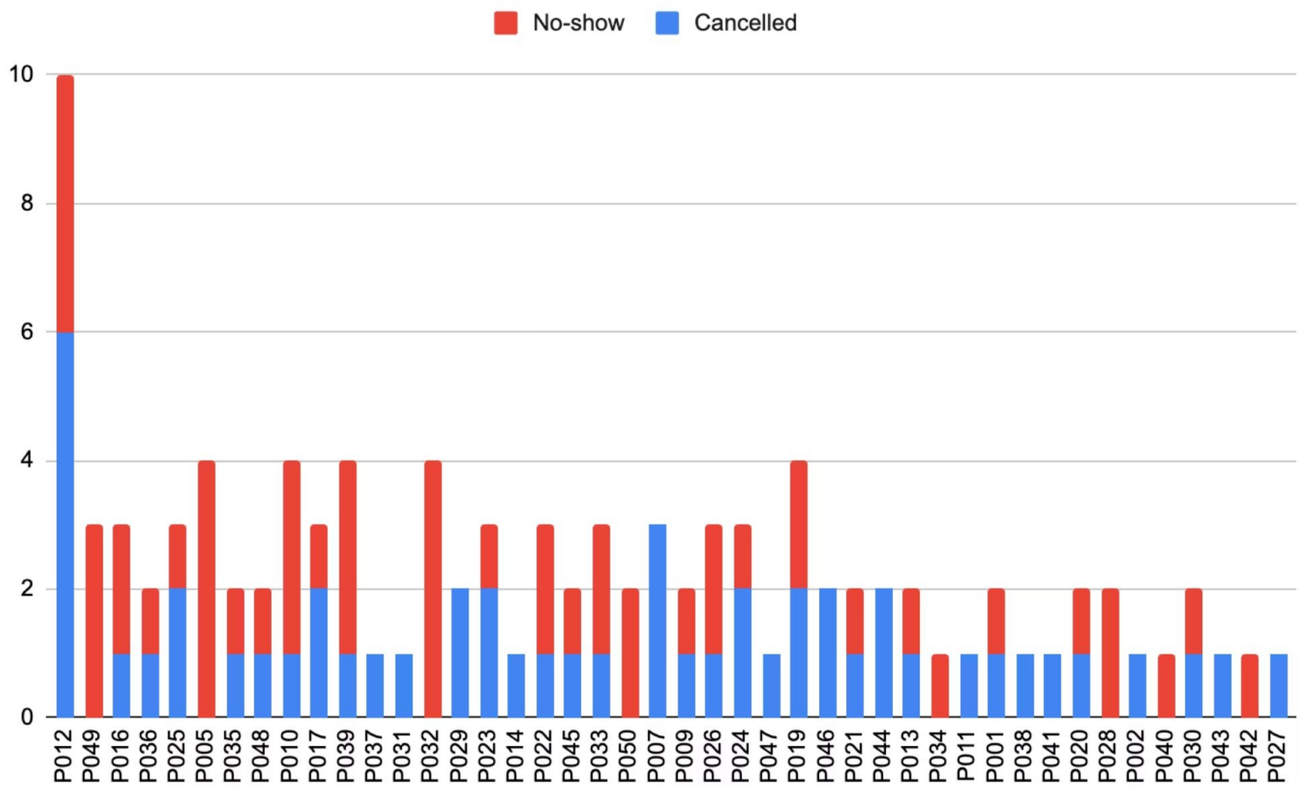
INSIGHT 3

In-Depth Patient Risk Analysis



Overlooking patient-related risks can lead to considerable financial and operational costs for the organisation.

A closer look at individual patient appointment histories revealed peculiar patterns. Patient **P012** stands out with ten appointments, none of which were completed: six were cancelled, and four were no-shows. This behaviour pattern indicates a high-risk patient who has consistently incurred costs without receiving services, suggesting a need for policy review regarding frequent cancellations or no-shows.

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Developing Robust Fraud Detection Logic

01

Anomaly Detection & Data Validation

Our initial step involved checking for anomalies and validating all data links across datasets. This ensured the integrity and accuracy of our source data, confirming the expected 200 appointment records were present and correctly linked.

02

Duplicate Record Identification

We then scanned for duplicate records, specifically targeting instances where the same email address was used for multiple patients. This helps identify potential identity fraud or data entry errors that could skew analysis and lead to incorrect billing.

03

Unfulfilled Appointment Payment Review

Building on earlier insights, we re-examined payments made for appointments that were either canceled or marked as no-shows. This crucial step directly flags instances where services were paid for but not rendered, indicating a potential area of systemic billing error or fraud.

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Actionable Recommendations for Claims Integrity

Enforce Data Accuracy & Uniqueness

To prevent deviations in records, it's crucial that all data entered into the dataset is accurate and free from duplicates. Our analysis highlighted cases where a single email address was associated with multiple patients, necessitating better data validation protocols.

Implement Robust Data Validation

Integrate rigorous data validation processes to ensure that all information is accurate, reliable, and consistent before it is processed. This includes applying unique constraints on columns that should not contain repeated values, such as email addresses.

Payment-Appointment Reconciliation Automation

Develop automated systems to cross-reference payment statuses with appointment completion. This will prevent payments from being issued for cancelled or no-show appointments, directly addressing the significant revenue loss identified in this report.

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